THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 7: **How do Organisms Reproduce?** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Asexual reproduction by budding occurs in: 1

(a) Planaria **(b)** Hydra **(c)** Plasmodium **(d)** Amoeba

2. The male reproductive part of a flower is the: 1

(a) sepal **(b)** petal **(c)** stamen **(d)** pistil

3. The site of fertilisation in human females is the: 1

(a) uterus **(b)** ovary

(c) vagina **(d)** fallopian tube (oviduct)

4. A student states: “Variation is created by asexual reproduction.” The correct response is: 1

(a) Incorrect — it creates very little variation

(b) Correct, it creates large variation

(c) Correct only in plants

(d) Cannot be determined

5. For Plasmodium, Student A says it reproduces by multiple fission; Student B says by budding. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. Rhizopus reproduces by: 1

(a) budding **(b)** fragmentation **(c)** spore formation **(d)** binary fission

7. The method used for plants like banana, rose and orange which have lost the capacity to produce seeds is: 1

(a) budding **(b)** vegetative propagation

(c) spore formation **(d)** fission

8. Which contraceptive method also protects against STDs? 1

(a) vasectomy **(b)** condom **(c)** oral pills **(d)** copper-T

9. The structure in which nutrition of the embryo occurs from the mother is: 1

(a) fallopian tube **(b)** placenta **(c)** ovary **(d)** cervix

10. The number of chromosomes in a human gamete is: 1

(a) 22 **(b)** 46 **(c)** 23 **(d)** 44

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is DNA copying essential for reproduction? 2

12. Name two STDs caused by bacteria. 2

13. What is the role of seminal vesicles and the prostate gland? 2

14. A student says that self-pollination and cross-pollination are the same. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe regeneration in Planaria. How is it different from fragmentation? 3

16. What changes are seen in girls at the time of puberty? 3

17. Draw a labelled diagram (describe) of the longitudinal section of a flower. 3

OR

Explain the process of fertilisation in flowering plants. 3

18. What are the advantages of vegetative propagation? (three) 3

OR

Why is family planning necessary? List two methods of birth control. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Draw/describe the human female reproductive system. 5
(ii) What happens to the egg if it is not fertilised?
(iii) How does the embryo get nourishment?

OR

(i) Describe the human male reproductive system. 5
(ii) Why are testes located outside the abdominal cavity?
(iii) Function of vas deferens.

20. **Case study:** In a lab, a student observed a slide of yeast showing small outgrowths and another of Rhizopus on bread showing black blobs. 5
(i) Name the methods of reproduction.
(ii) What are the black blobs?
(iii) How do spores survive harsh conditions?
(iv) Name one more organism reproducing by budding.

OR

(i) Why is sexual reproduction important for variation? 5
(ii) Explain the significance of half the number of chromosomes in gametes.
(iii) Compare asexual and sexual reproduction (two points).

END OF QUESTION PAPER — Total: 40 Marks